

Appendix: Exact Maximum Flexi-Clique Computation under Non-Hereditary Feasibility and Connectivity Constraints

A Hardness proof of maximum Flexi-clique problem

This section establishes the computational hardness of the Maximum Flexi-clique problem in two steps. Appendix A.1 proves NP-hardness by a reduction from the k -clique problem, and Appendix A.2 strengthens this result by showing that the problem admits no constant-factor approximation unless $P = NP$.

A.1 NP-hardness

LEMMA 1. *The Flexi-clique decision problem is NP-hard.*

PROOF. We prove NP-hardness by reduction from the well-known k -clique problem, which is NP-complete in its decision form. Given an instance (G', k) of the clique decision problem, we construct an equivalent instance (G, τ, k) of the Flexi-clique decision problem as follows. We set $G = G'$, meaning that the node and edge sets remain unchanged. We then choose $\tau \in (0, 1)$ sufficiently close to 1 so that for any subset of size x , the degree threshold satisfies $\lfloor x^\tau \rfloor = x - 1$. Intuitively, this setting makes the Flexi-clique condition equivalent to the clique constraint. Formally, it suffices to select τ satisfying

$$\frac{\ln(|V| - 1)}{\ln(|V|)} < \tau < 1,$$

since for any $x < x'$, the ratio $\frac{\ln(x-1)}{\ln(x)} < \frac{\ln(x'-1)}{\ln(x')}$. The parameter k remains the same. If G' contains a clique H of size at least k , then H also forms a valid Flexi-clique in G , because each node has degree $|H| - 1$ within H . Conversely, if G contains a Flexi-clique H of size at least k , the degree constraint forces H to be a clique, implying that G' contains a clique of size at least k . Since the transformation runs in polynomial time, the Flexi-clique decision problem is NP-hard. $\square$

LEMMA 2. *The Maximum Flexi-clique problem is NP-hard.*

PROOF. We prove NP-hardness by reducing from the Flexi-clique decision problem (shown to be NP-hard in Lemma 1). Suppose we have a polynomial-time algorithm $\mathcal{A}$ that finds the maximum Flexi-clique in any instance (G, τ) . Given an instance (G, τ, k) of the k -Flexi-clique problem, we:

- (1) Use $\mathcal{A}$ to compute the largest Flexi-clique $H_{\max}$ and its size $|H_{\max}|$.
- (2) If $|H_{\max}| \geq k$, return yes; otherwise, return no.

Since it solves the NP-hard k -Flexi-clique problem in polynomial time, the Maximum Flexi-clique problem must also be NP-hard. $\square$

A.2 Constant factor inapproximability

THEOREM 1. *The Maximum Flexi-clique problem is constant-factor inapproximable unless $P = NP$.*

PROOF. We prove the claim via a gap-preserving reduction from the gap version of the CLIQUE problem. Specifically, given a graph $G = (V, E)$, an integer K , and a constant $c > 1$, the task is to distinguish between the following two cases, under the promise that one of them holds:

- **YES:** G contains a clique of size at least K ;
- **NO:** every clique in G has size at most K/c .

It is NP-hard to distinguish between the above two cases [1]. Given a graph $G = (V, E)$ with parameters K and $c > 1$, under the promise that G contains a clique of size at least K or that every clique in G has size at most K/c , we construct a graph $G' = (V', E')$ that preserves the promised gap while amplifying differences in clique size. Specifically, for each node $v \in V$, we create a clique C_v consisting of exactly B nodes, where B is a sufficiently large polynomial in n . The node set of G' is defined as $V' = \bigcup_{v \in V} C_v$, and we denote $|V'|$ by N .

For any two distinct nodes $u, v \in V$, the cliques C_u and C_v are made fully adjacent if $(u, v) \in E$, and completely non-adjacent otherwise.

Next, we set the parameter τ as

$$\tau = 1 - \frac{1}{4n \log N},$$

which will be used to control the gap between s and s^τ in the analysis below. The graph construction process for the gap-preserving reduction is illustrated in Figure 1. For the constructed graph G' , we denote by $\text{OPT}(G')$ the maximum size of a Flexi-clique.

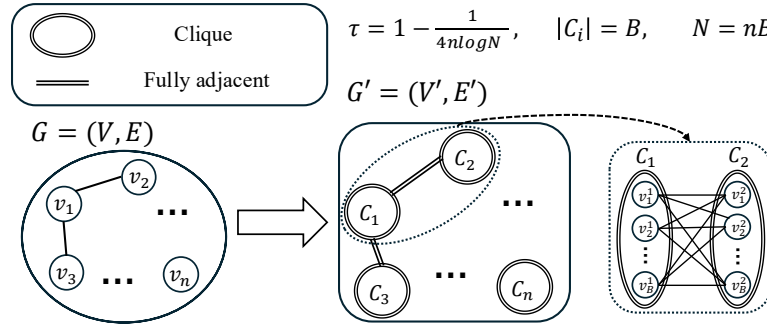


Fig. 1. Reduction process.

YES Case. Assume that G contains a clique $Q \subseteq V$ with $|Q| = K$, and let $S = \bigcup_{v \in Q} C_v \subseteq V'$. Clearly, the set S has size KB , and the induced subgraph $G'[S]$ is connected since all cliques corresponding to nodes in Q are adjacent. Consider any node $x \in S$, since C_v is a clique, x has $B - 1$ neighbours inside $G'[C_v]$. Moreover, from the other $K - 1$ cliques in S , the node x has in total $(K - 1)B$ neighbours. Therefore,

$$\deg_{G'[S]}(x) \geq (B - 1) + (K - 1)B = KB - 1. \quad (1)$$

As $\tau \in (0, 1)$, $KB - 1 \geq \lfloor (KB)^\tau \rfloor$ for the chosen value of B . Hence $\deg_{G'[S]}(x) \geq \lfloor |S|^\tau \rfloor$ for all $x \in S$, and thus S is a valid Flexi-clique with $|S| = KB$, implying

$$\text{OPT}(G') \geq KB. \quad (2)$$

NO Case. Now assume that every clique in G has size at most K/c . Let $S \subseteq V'$ be an arbitrary **Flexi-clique** in G' . For each $v \in V$, define $a_v = |S \cap C_v|$, and let

$$U = \{v \in V : a_v > 0\}. \quad (3)$$

Denote

$$s = |S| = \sum_{v \in U} a_v. \quad (4)$$

We first bound the total number of non-neighbours that a node in a **Flexi-clique** can have. Recall that $\tau = 1 - \frac{1}{4n \log N}$ and that $s \leq nB = N$. Thus,

$$s - s^\tau \leq \frac{s}{4n} \leq \frac{B}{4}. \quad (5)$$

Fix any $u \in U$ and consider an arbitrary node $x \in S \cap C_u$. By construction of G' , the degree of x in the induced subgraph $G'[S]$ is

$$\deg_{G'[S]}(x) = (a_u - 1) + \sum_{\substack{v \in U: \\ (u,v) \in E}} a_v. \quad (6)$$

Therefore, the number of nodes in S that are non-adjacent to x is

$$(s - 1) - \deg_{G'[S]}(x) = \sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v. \quad (7)$$

Since S is a **Flexi-clique**, we have $\deg_{G'[S]}(x) \geq \lfloor s^\tau \rfloor$, and hence

$$\sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v \leq (s - 1) - \lfloor s^\tau \rfloor \leq s - s^\tau \leq \frac{B}{4}, \quad (8)$$

where the last inequality follows from (5).

Now let C be a maximum clique in the induced graph $G[U]$. By the NO-case assumption, we have $|C| \leq K/c$. By maximality of C , for every $v \in U \setminus C$ there exists a node $u \in C$ such that $(u, v) \notin E$, and we assign each $v \in U \setminus C$ to one such u .

For each $u \in C$, let $A(u)$ denote the set of nodes in $U \setminus C$ assigned to u , that is,

$$A(u) = \{v \in U \setminus C : v \text{ is assigned to } u\}.$$

Then the sets $\{A(u) : u \in C\}$ form a partition of $U \setminus C$. Moreover, by the assignment rule, every node $v \in A(u)$ satisfies $(u, v) \notin E$. Hence, for every fixed $u \in C$,

$$A(u) \subseteq \{v \in U : (u, v) \notin E\}.$$

It follows from (8) that

$$\sum_{v \in A(u)} a_v \leq \sum_{\substack{v \in U: \\ (u,v) \notin E}} a_v \leq \frac{B}{4}. \quad (9)$$

Summing over all $u \in C$, we obtain

$$\sum_{v \in U \setminus C} a_v = \sum_{u \in C} \sum_{v \in A(u)} a_v \leq |C| \cdot \frac{B}{4}. \quad (10)$$

Since each block C_v has size exactly B , we also have

$$\sum_{u \in C} a_u \leq |C|B. \quad (11)$$

Combining this with (10), we conclude that

$$|S| = \sum_{u \in C} a_u + \sum_{v \in U \setminus C} a_v \leq |C|B + |C| \cdot \frac{B}{4} = \frac{5}{4}|C|B \leq \frac{5}{4} \cdot \frac{K}{c}B. \quad (12)$$

Therefore,

$$\text{OPT}(G') \leq \frac{5}{4} \cdot \frac{K}{c}B. \quad (13)$$

Combining (2) and (13) yields a gap of $\frac{KB}{(5/4)(K/c)B} = \frac{4c}{5}$ between the **YES** and **NO** instances. Since c can be chosen as an arbitrary constant, any constant-factor approximation for the Maximum Flexi-clique problem would distinguish the promised cases of CLIQUE, unless $P = NP$. □

B Complexity analysis

This section analyses the complexity of EBA in two parts. Appendix B.1 analyses the running time: it establishes the per-node cost and the worst-case bound $O^*(2^n)$, explains why—in contrast to the established cohesive subgraph models—no branching base strictly below 2 can be guaranteed for Flexi-clique, and then proves a conditional bound below 2^n under bounded branching width. Appendix B.2 shows that the Flexi-clique model admits no size threshold that guarantees connectivity. Throughout, $\tau \in (0, 1)$ is the degree parameter and $n = |V|$.

B.1 Running time analysis

Per-node cost and worst-case bound. By the completeness of CP-Branching and the soundness of the pruning rules established in the main paper, EBA computes a maximum Flexi-clique exactly. Its running time is the product of the number of search-tree nodes and the cost spent at each of them. Consider a search node $B = (S, C^r, C^{un}, D)$, and write $V_B = S \cup C^r \cup C^{un}$ for its remaining scope and E_B for the edge set of $G[V_B]$. The degree-based rule takes $O(|S| + |E_B|)$ time for the adjusted-degree computation and the follower cascade, the size-bound check takes $O(|S|)$ time, and the distance-based rule takes $O(|V_B| + |E_B|)$ time through incremental maintenance of distances. The work at each search node is therefore polynomial in $|V_B| + |E_B|$. Since every branch commits at least one node to either S or D , the search tree has $O(2^n)$ nodes, and the total running time is $O^*(2^n)$.

Why a base below 2 cannot be guaranteed. For the established cohesive subgraph models—clique, k -plex, and γ -quasi-clique—branching algorithms with a worst-case base strictly below 2 have all been proven [9, 11, 12]. The common reason is that each of these models keeps every node’s required degree close to the subgraph size, which limits how many non-neighbours a node may have. A clique requires degree $|H| - 1$ and thus no non-neighbour; a k -plex requires degree $|H| - k$ and thus at most $k - 1$ non-neighbours per node; and a γ -quasi-clique requires degree $\gamma(|H| - 1)$ and thus at most a $(1 - \gamma)$ fraction of non-neighbours. In every case, the number of non-neighbours a node may have—which is the source of the branching width—is held to a constant or to a fixed fraction of the subgraph size, so a characteristic-root analysis yields a base below 2.

Flexi-clique breaks this pattern. Its degree requirement $\lfloor |H|^\tau \rfloor$ is sublinear, falling ever further behind $|H|$ as the subgraph grows, so a node may be non-adjacent to almost the entire subgraph: the number of non-neighbours a node may

have grows with $|H|$ rather than staying a constant or a fixed fraction below it. Since this quantity is exactly the source of the branching width, the width can grow with the subgraph size, and no branching scheme can guarantee a base below 2 in the worst case. This intrinsic gap is why exact **Flexi-clique** computation is harder than for the established models, and why the graph scales and running times that **EBA** handles cannot be placed on the same footing as those of algorithms tailored to them. On real instances, however, the width-based pivot keeps the branching width small; the remainder of this section quantifies the resulting gain, and Appendix C.1 measures the width empirically.

A conditional bound below 2^n . Recall that **EBA** explores a branch-and-bound tree whose search nodes have the form $B = (S, C^r, C^{un}, D)$. As the progress measure of a search node we use the number of undecided candidates, $\mu(B) = |C^r| + |C^{un}|$. In CP-Branching, the child B_i commits the candidates $v_1, \dots, v_i$ to S or D , and therefore reduces μ by at least i ; equivalently, if the explored children are ordered by rank, the j -th of them reduces μ by at least j . The following theorem shows that whenever the width-based pivot keeps the branching narrow, the base of the exponential drops strictly below 2.

THEOREM 2. *Suppose the width-based pivot bounds the number of explored children at every search node to at most w . Then **EBA** runs in $O^*(\alpha_w^n)$, where α_w is the largest root of*

$$x^{w+1} - 2x^w + 1 = 0, \quad (14)$$

and $\alpha_w < 2$ for every $w \geq 2$.

PROOF. Let $T(\mu)$ bound the size of the subtree rooted at a search node of measure μ . Since at most w children are explored and the j -th of them reduces μ by at least j , we obtain the recurrence $T(\mu) \leq \sum_{j=1}^w T(\mu - j)$. Substituting $T(\mu) = \alpha^\mu$ and dividing by $\alpha^{\mu-w}$ gives $\alpha^w = \sum_{l=0}^{w-1} \alpha^l = (\alpha^w - 1)/(\alpha - 1)$, and multiplying both sides by $(\alpha - 1)$ yields the characteristic equation (14). It remains to locate its largest root. Write $f(x) = x^{w+1} - 2x^w + 1$. Then $f(1) = 0$ and $f(2) = 2^{w+1} - 2 \cdot 2^w + 1 = 1 > 0$, while f attains a negative local minimum at $x = 2w/(w+1) < 2$ for every $w \geq 2$; hence the largest root α_w lies in the open interval $(1, 2)$. Since the per-node work is polynomial in the size of G , the total running time is $T(n) = O^*(\alpha_w^n)$. $\square$

For $w = 2, 3, 4$, the characteristic roots are $\alpha_w \approx 1.618, 1.839, 1.928$, respectively, and α_w approaches 2 as w grows. Keeping the width small is therefore what makes the bound meaningful, and this is precisely what the width-based pivot targets.

How the pivot bounds the width. The pivot selects the node p with the smallest margin $\bar{d}_{\text{margin}}(p)$ —the number of further non-neighbours p can tolerate before violating its degree constraint—and orders the non-neighbours of p first over the whole reachable set. Each admitted non-neighbour raises the in- S non-degree of p by one, so after $\bar{d}_{\text{margin}}(p)$ admissions p saturates, and every later child—including the include-all child, whose partial set then contains all non-neighbours of p —is pruned. The number of explored children is thereby held to about $\bar{d}_{\text{margin}}(p) + 1$; that is, w is on the order of the pivot’s margin. On instances where this margin remains small, the base is well below 2. As discussed above, the sublinear degree requirement of **Flexi-clique** prevents our width-based analysis from guaranteeing a constant w in the worst case; Appendix C.1 shows that the observed width nevertheless remains small in practice.

B.2 Absence of a size threshold for connectivity

Classical cohesive subgraph models either guarantee connectivity outright or recover it beyond a modest threshold: a clique is trivially connected; a k -plex H is connected—with diameter at most two—whenever $|H| \geq 2k - 1$ [8]; and a

γ -quasi-clique is connected whenever $\gamma \geq 0.5$ [7, 12]. The **Flexi-clique** model, in contrast, imposes only a size-dependent degree constraint, and no analogous threshold exists: arbitrarily large node sets can satisfy the degree constraint while remaining disconnected. We make this precise below.

PROPERTY 1 (NO SIZE THRESHOLD FOR CONNECTIVITY). *For any $\tau \in (0, 1)$ and any integer h , there exists a node set $H \subseteq V$ with $|H| \geq h$ such that every node $v \in H$ satisfies $d(v, G[H]) \geq \lfloor |H|^\tau \rfloor$, yet $G[H]$ is disconnected.*

PROOF. Fix any $\tau \in (0, 1)$ and any integer h . Choose an integer s such that

$$2s \geq h \quad \text{and} \quad s - 1 \geq \lfloor (2s)^\tau \rfloor. \quad (15)$$

Such an s always exists for sufficiently large s , since $(2s)^\tau$ grows strictly sub-linearly in s for $\tau < 1$. Now construct a graph $G = (V, E)$ consisting of two disjoint cliques C_A and C_B , each of size s , with no edges between them, and let $H = C_A \cup C_B$, so that $|H| = 2s \geq h$. Each node $v \in H$ has degree $s - 1$ in $G[H]$, and by (15) we have $s - 1 \geq \lfloor (2s)^\tau \rfloor = \lfloor |H|^\tau \rfloor$; hence every node satisfies the **Flexi-clique** degree constraint. However, since there is no edge between C_A and C_B , the induced subgraph $G[H]$ is disconnected. As h was arbitrary, the **Flexi-clique** model admits no size-based threshold that guarantees connectivity. $\square$

Consequently, connectivity cannot be inferred from the degree constraint and must be enforced explicitly during the search, which motivates the connectivity-preserving branching of **EBA**.

C Additional experiments

This section reports additional experiments: an empirical measurement of the branching width (Appendix C.1), an ablation of the connectivity-preserving branching scheme (Appendix C.2), a comparison with the exact maximum quasi-clique solver **IterQC** (Appendix C.3), and a case study on density distributions (Appendix C.4).

Table 1. Per-node branching width on LFR graphs S_3 – S_6 as the graph size n grows ($\tau=0.7$).

Graph	S_3	S_4	S_5	S_6
n	10^3	10^4	10^5	10^6
W_{med}	3	3	2	2
W_{max}	6	17	11	8
$\alpha_{W_{\text{med}}}$	1.839	1.839	1.618	1.618

Table 2. Branching width on real networks ($\tau \in \{0.5, 0.7\}$); W_{max}/n stays below 1%.

Dataset	AP	CD	CO	GW	DB	FD
n	16,046	39,577	90,303	196,591	317,080	1,070,376
ans.	252	360	255	1443	899	7
W_{med}	44	231	20	255	34	3
W_{max}/n	0.00798	0.00644	0.00194	0.00130	0.00080	3.7×10^{-6}

C.1 Empirical branching width

Theorem 2 shows that if the width-based pivot keeps the number of explored children at every search node bounded by a constant w , then EBA runs in $O^*(\alpha_w^n)$ with $\alpha_w < 2$. The bound is conditional: as discussed in Appendix B.1, the sublinear degree requirement of Flexi-clique prevents any constant w from being guaranteed in the worst case. We therefore measure the branching width empirically, instrumenting EBA to record, at every CP-Branching search node, the number of children it actually explores, and report the median width W_{med} and the maximum width W_{max} . The measurement uses two families of graphs: LFR synthetic networks [4] S_3 – S_6 , generated with mixing parameter $\mu = 0.3$, maximum degree 50, and maximum community size 10^3 , varying only the number of nodes so that S_i denotes the instance with $n = 10^i$ nodes for $i \in \{3, \dots, 6\}$; and the real networks in Table 2.

The key question is whether the width stays small relative to the graph size, and two findings show that it does. First, the width does not scale with n : as Table 1 reports, the median width on S_3 – S_6 stays at 2–3 while the number of nodes grows from 10^3 to 10^6 , corresponding to an effective base between $\alpha_2=1.618$ and $\alpha_3=1.839$. Second, on real networks (Table 2), the width remains a tiny fraction of n : the maximum width is at most a few hundred even on graphs with hundreds of thousands to millions of nodes. This is because w tracks the size of the Flexi-clique itself, which is governed by the degree structure of the graph rather than by its size. Thus, while a constant worst-case width cannot be guaranteed, in practice the branching width stays orders of magnitude below n and does not grow with it, so EBA effectively explores a search tree far narrower than the trivial 2^n bound.

Table 3. Running time (s) of CP-Branching and SE under identical pruning ($\tau=0.5$).

Scheme	EM	ED	FD	PG	AP	CD	CO	GW	DB
CP-Branching	0.02	0.02	1.31	0.05	0.67	1.90	1.64	47.8	23.5
SE	0.04	0.03	1.20	T/O	T/O	T/O	T/O	T/O	T/O

C.2 Contribution of connectivity-preserving branching

To isolate the contribution of CP-Branching, we compare it against SE, a set-enumeration branching scheme in the style of quasi-clique enumerators that includes each candidate in turn and recurses on the strictly later ones. Both schemes explore the same solution space under identical pruning rules, with τ fixed to 0.5; the only difference is the branching. Since SE does not distinguish candidates by reachability, it may extend a partial set with any remaining candidate, and therefore has to verify connectivity, on top of the degree constraints, for every partial set it forms; CP-Branching instead maintains connectivity by construction and never performs an explicit check.

Table 3 reports the running time. When the search space remaining after the initial pruning is narrow, the two schemes behave almost identically. Once substantial branching is required, however, the undifferentiated candidate set of SE makes its branches grow excessively long while the connectivity checks accumulate, and SE fails to terminate within the 10-minute budget on all six remaining datasets, whereas CP-Branching finishes each of them within a minute. This asymmetry explains why set-enumeration branching serves the previous models well—their hereditary structure or bounded diameter renders connectivity essentially free—but breaks down for Flexi-clique, which satisfies neither property (Appendix B.2); explicitly separating candidates by reachability, as CP-Branching does, is what keeps exact search tractable.

Table 4. Running time (s) of EBA and IterQC.

Alg.	τ	EM	ED	PG	AP	DI	CA	CD	BK	CR	CO	FC	GW	DB	FD	YT
EBA	0.5	0.02	0.03	0.05	0.73	0.10	0.11	4.19	1.26	3.23	1.97	45.92	45.08	22.93	1.47	55.75
	0.7	0.03	0.03	0.03	0.98	0.07	0.08	0.29	0.51	1.94	0.30	26.65	2.93	3.15	1.58	T/O
IterQC	0.5	0.04	0.03	2.10	0.52	0.32	1.02	0.55	0.65	T/O	0.26	4.03	1007.2	1.15	1.18	7034.4
	0.7	0.09	0.05	0.18	0.56	0.32	0.83	0.55	0.65	3.08	0.25	32.69	0.98	0.60	1.32	4.38

Table 5. Sizes of the maximum Flexi-clique and the maximum quasi-clique.

Alg.	τ	EM	ED	PG	AP	DI	CA	CD	BK	CR	CO	FC	GW	DB	FD	YT
EBA	0.5	440	80	194	993	195	168	360	561	1370	255	9024	1443	899	15	2208
	0.7	137	30	116	252	88	77	93	208	460	87	1846	270	259	7	T/O
IterQC	0.5	67	21	49	107	51	45	49	101	T/O	45	1051	87	115	7	75
	0.7	42	12	42	62	35	28	31	72	82	43	576	51	114	5	39

C.3 Comparison with exact maximum γ -quasi-clique search

The closest exact baseline of EBA is IterQC [10], a state-of-the-art maximum γ -quasi-clique solver. Both target non-hereditary cohesive models, but the quasi-clique problem is structurally easier in two respects. First, its degree threshold grows linearly with the subgraph size, whereas the Flexi-clique threshold $\lfloor n^\tau \rfloor$ is sublinear (Appendix B.1). Second, for $\gamma \geq 0.5$ a quasi-clique is guaranteed to be connected with diameter at most two [7], so IterQC can restrict the search to local, bounded-diameter neighbourhoods and never needs an explicit connectivity mechanism; Flexi-clique offers no such guarantee (Appendix B.2), which is why EBA relies on connectivity-preserving branching. Moreover, IterQC does not even apply for $\gamma < 0.5$, whereas EBA solves the full range of τ .

Tables 4 and 5 compare EBA (τ) against IterQC (γ) at the matched parameter values 0.5 and 0.7 under a 1-day time budget. Two observations stand out. First, despite solving the harder problem, EBA is competitive and often faster. Second, the maximum Flexi-clique is substantially larger than the maximum quasi-clique at the same parameter, so EBA attains comparable running time while certifying optimality over a much larger and harder search space.

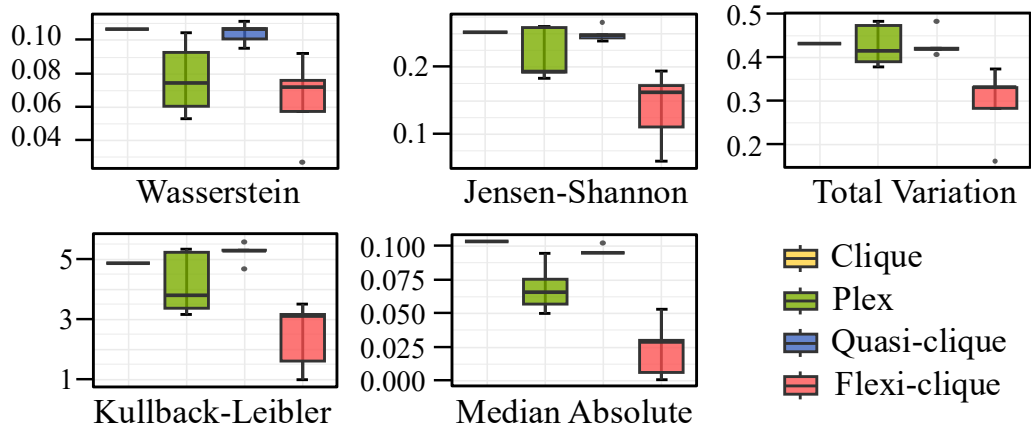


Fig. 2. Density distribution distance comparison among cohesive subgraph models.

C.4 Case study: density distribution comparison

To provide a qualitative case study on how different cohesive subgraph models reflect observed density variation, we use the Eu-mail network [6], a temporal email graph. For each time window i , we construct a snapshot graph G_i from the email interactions within that period, yielding a sequence of N graphs $G_1, \dots, G_N$. The density distribution of these temporal snapshots is used as an empirical reference pattern for the interaction structures observed over time. For each G_i , we extract the maximum subgraph H_i under each of four models and record its density $\frac{2|E(H_i)|}{|H_i|(|H_i|-1)}$, where $E(H_i)$ denotes the edge set of the induced subgraph $G_i[H_i]$. We then compare the resulting density distributions across all snapshots with the empirical reference using five distance metrics: Wasserstein distance [2], Jensen–Shannon divergence [5], Total Variation distance, symmetric Kullback–Leibler divergence [3], and Median Absolute Difference. The models and parameter settings are as follows: maximum clique; k -plex with $k \in \{2, 4, 6, 8, 10\}$; γ -quasi-clique with $\gamma \in \{0.5, 0.6, 0.7, 0.8, 0.9\}$; and Flexi-clique with $\tau \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$. Note that $k = 1$ is excluded because it reduces to the clique model, and $\gamma < 0.5$ is excluded because exact solutions cannot be guaranteed in that range. For all metrics, smaller values indicate closer agreement with the empirical reference distribution.

Figure 2 summarises the results. The clique model performs poorly across all metrics because it always returns density 1.0, and therefore cannot reflect variation across snapshots. The k -plex model provides a more flexible alternative, but its density distributions still remain noticeably different from the empirical reference. The γ -quasi-clique model also shows limited agreement, likely because its fixed threshold does not adapt to variation across different snapshot sizes. Among the compared models, Flexi-clique yields the closest match to the empirical density distribution across all five metrics. This suggests that the size-adaptive degree constraint of Flexi-clique better mirrors the density variability observed in this temporal interaction setting.

References

- [1] Uriel Feige, Shafi Goldwasser, László Lovász, Shmuel Safra, and Mario Szegedy. 1991. Approximating clique is almost NP-complete. In *[1991] Proceedings 32nd Annual Symposium of Foundations of Computer Science*. IEEE Computer Society, 2–12.
- [2] Leonid V Kantorovich. 1942. On the translocation of masses. In *Dokl. Akad. Nauk. USSR (NS)*, Vol. 37, 199–201.
- [3] Solomon Kullback and Richard A Leibler. 1951. On information and sufficiency. *The annals of mathematical statistics* 22, 1 (1951), 79–86.
- [4] Andrea Lancichinetti, Santo Fortunato, and Filippo Radicchi. 2008. Benchmark graphs for testing community detection algorithms. *Physical review E* 78, 4 (2008), 046110.
- [5] Jianhua Lin. 2002. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information theory* 37, 1 (2002), 145–151.
- [6] Ashwin Paranjape, Austin R Benson, and Jure Leskovec. 2017. Motifs in temporal networks. In *Proceedings of the tenth ACM international conference on web search and data mining*. 601–610.
- [7] Jeffrey Pattillo, Alexander Veremyev, Sergiy Butenko, and Vladimir Boginski. 2013. On the maximum quasi-clique problem. *Discrete Applied Mathematics* 161, 1-2 (2013), 244–257.
- [8] Stephen B Seidman and Brian L Foster. 1978. A graph-theoretic generalization of the clique concept. *Journal of Mathematical sociology* 6, 1 (1978), 139–154.
- [9] Etsuji Tomita, Akira Tanaka, and Haruhisa Takahashi. 2006. The worst-case time complexity for generating all maximal cliques and computational experiments. *Theoretical computer science* 363, 1 (2006), 28–42.
- [10] Hongbo Xia, Kaiqiang Yu, Shengxin Liu, Cheng Long, and Xun Zhou. 2025. Maximum Degree-Based Quasi-Clique Search via an Iterative Framework. In *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining V. 2*. 3285–3296.
- [11] Mingyu Xiao, Weibo Lin, Yuanshun Dai, and Yifeng Zeng. 2017. A fast algorithm to compute maximum k -plexes in social network analysis. In *Proceedings of the AAAI conference on Artificial Intelligence*, Vol. 31.
- [12] Kaiqiang Yu and Cheng Long. 2023. Fast maximal quasi-clique enumeration: A pruning and branching co-design approach. *Proceedings of the ACM on Management of Data* 1, 3 (2023), 1–26.